Tax Workpaper Case Study — Submission

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The prototype

The workpaper is a web app at Okafor-Lindqvist Workpaper. It loads the packet exactly as delivered (`packet.json` , unmodified) and builds everything else in the browser.

- **Role switcher** (top bar, "Acting as"): all six staff plus both clients. Permissions are enforced. Marcus sees masked SSNs. Sam can't act before 2027-01-04. Clients get a portal with their questions and the 8879 approval.
- **Season clock** (top bar, "As of"): replays the engagement from the first uploads in January 2026 through the e-file reject and acceptance to the IRS Letter 566 in January 2028. Documents, engine versions, messages and agency records appear on the dates they arrived.
- **Saving**: every action goes to an append-only journal. Inside claude.ai that journal is a shared database with live updates, so several people can work at once. Anywhere else it falls back to browser storage.
- **Demo data**: Firm › "Load demo season activity" replays about 70 scripted actions from March to July 2026. They are labelled as demo data. "Reset workpaper journal" clears them.
- **Source**: Preact + htm with no build framework, about 3,000 lines across `model.js` (evidence graph), `checks.js` (independent checks), `journal.js` (events, replay, permissions) and the views. *(Add your GitHub link here.)*

> Before you send the link, share the artifact (Share menu) so reviewers outside your account can open it.

Core idea and data model

A workpaper is an evidence graph with a ledger. A spreadsheet workpaper copies numbers from tab to tab and asks people to draw tickmarks next to them. This one keeps two things apart: the facts the pipeline delivered, which nobody edits, and an append-only journal of what people did. The workpaper on screen is a replay of the facts plus the journal up to a point in time.

Facts: the evidence graph. The packet loads as delivered and becomes a graph:

- *Return line* (per engine version) → its `inputs` (how it was built) and its `sources` (where it came from).
- *Extracted value* → a field on a page of a document, with what was read and the confidence.
- *Document* → every line it reached, or a status saying why it reached none: superseded, duplicate image, reference only, support only, or not in this version.
- Client answers, messages, call notes, prior-year pointers, agency records, assets and engine parameters are nodes too. Every reference format in the README resolves to something you can open.

Indexes run both ways (line → source, source → lines, line → downstream lines), so a number traces down to the printed page and a document traces up to the return.

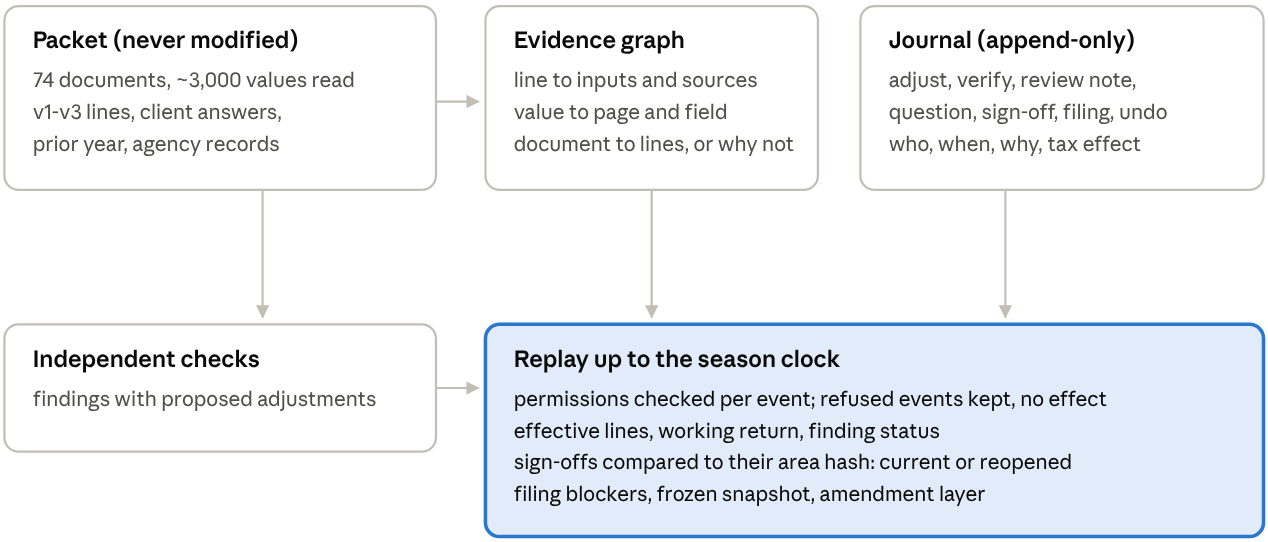
Work: the journal. Each action is an immutable event `{id, at, actor, role, type, target, data, batch}`: adjust, verify, note, review note, reply, clear, question, answer, decision, sign-off, attach, time, unmask, filing step, undo. Nothing is overwritten. One choice gives most of the brief:

- **Originals kept.** An adjustment stores from/to per line, who, why, the finding it resolves and its tax effect. A revert is another event.
- **Enforced roles.** Permissions are checked during replay, not only in the UI. A forbidden event is recorded, shown as refused with the reason, and has no effect. Rules enforced: role capabilities, no self-review, partner approval above 1,000 of tax effect, staff active dates, unmasking only with clearance, every judgment needs an authority, and no e-file while review notes, unapproved adjustments, error findings or stale approvals remain.
- **Sign-offs that reopen.** A sign-off stores a hash of every number, adjustment and approval in its area. When anything changes, including a new engine version, the sign-off shows as reopened. The client's 8879 approval fingerprints the bottom line the same way.
- **History for free.** "Since I last looked", version comparison (v1 / v2 / v3 / working / filed), undo of an action or a batch, freeze at filing with an amendment layer on top, and replaying the season.

Tax effect without an engine. Each adjustment carries its effect, computed from the version's per-dollar rates or taken from the engine's exact what-if where one exists. Lines downstream of an adjustment are marked "recompute" rather than re-added by hand,

because the real system recomputes.

Facts and work stay separate; the workpaper is a replay of both



data model · facts, journal, replay

People only ever add events; the screen is always recomputed from the packet and the journal.

Errors found and how the design surfaced them

Twenty errors survive in engine v3. Fixing them turns a federal balance due of 4,220 into an estimated refund of 4,680 and raises the California refund from 8,256 to 11,627. Federal tax falls 9,900 and California tax falls 6,871. Federal payments drop 1,000 and California payments drop 3,500. Every one came from a check that reads the source documents, agency records, client answers or the prior year, never from the engine's own diagnostics. Effects use the engine's per-dollar rates, or the exact what-if where the engine supplied one. Positive numbers raise tax.

#	Error	Type	How it surfaced	Fed tax	CA tax
1	Three noncovered RSU lots sold with 0 basis; supplemental statement D11 gives 19,100 / 19,100 / 9,420 (vest value already in W-2 wages)	Engine	Each noncovered lot matched to a supplement lot by acquisition date	−15,536	−2,765

#	Error	Type	How it surfaced	Fed tax	CA tax
2	Paywell 1099-K (12,400) added on top of books revenue; it is Atlas's card payments, already in books and in Atlas's 1099-NEC	Engine	Books reconciled to all third-party returns (D15 p3)	−4,402	−384
3	Halvorsen wages after the move treated as California-source (box 16 kept CA through 09/30); paystub D73 gives 22,469 through 07/15	Engine	Wages split by residency dates and paystub YTD, not box 16	—	−5,685
4	Self-employed health insurance 14,400 deducted although Erik was eligible for Ngozi's employer plan (qa-25; same call on the 2024 return)	Engine	Questionnaire + prior-year note tested against the rule	+3,571	+605
5	Whole HVAC system replacement (new heat pump, permit) expensed as a repair	Engine	Invoice read against capitalization rules; exact engine what-if	+3,633	+896
6	Seller's share of 2025 property tax (6,912 credit at closing) included in the buyers' SALT; drops SALT below the cap	Engine	Deduction tied to the buyer closing statement	+1,462	+332
7	Crypto lot C6 at 0 basis and short-term; client says bought 03/15/2021 for 3,050 (bank statement still needed)	Judgment	Engine diagnostic + client message, routed as a question	−1,312	−195
8	Mileage log page 3 is a second scan of page 2: 11,800 miles claimed, 8,100 real	Engine	Page contents compared; trips summed per page	+919	+80
9	Car donation at Blue Book 6,000; charity sold it for 2,900 (1098-C box 4c)	Engine	Claimed value compared with the 1098-C	+806	+149
10	Same bank-sweep interest (1,840) from the composite 1099 and the sweep bank's 1099-INT	Engine	Account numbers matched across documents; 2024 showed it once	−640	−127

#	Error	Type	How it surfaced	Fed tax	CA tax
11	Payoff interest (2,150) added again although the 1098 says box 1 includes it	Engine	Lender statements' own notes	+559	+103
12	SEP limit not recomputed after the Schedule C fixes (15,214 → 13,674)	Engine	Limit recomputed from adjusted profit	+477	+65
13	Family phone plan and home internet at 100%; client gave 70% / 50%	Judgment	Books-to-tax review against call notes	+464	+41
14	Personal Spotify Family (216) in business software	Engine	Transaction detail against call notes	+77	+7
15	Swedish interest converted at the 2024 average rate (10.56) instead of 2025 (9.86)	Engine	Rate key in the line's sources checked against tax year	+43	+9
16	Meridian page 32 never read: lot M18 missing (explains the 250 summary gap)	Extraction	Unread page parsed to explain the engine's mismatch diagnostic	−21	−2
17	Q4 federal estimate was 3,000 (IRS transcript), not the 4,000 the client remembered	Client data	Payments agreed to the IRS transcript	payments −1,000	—
18	09/15/2025 Web Pay 3,500 designated to 2024; FTB holds it as a 2024 credit	Engine	Payments agreed to the FTB account	—	payments −3,500
19	Theo's SSN transposed in the organizer (…3309 vs … 3390 on the 2024 return and SS card)	Client data	Identity agreed to the prior year; flagged before the 07/13 e-file reject	—	—
20	Little Sprouts tax ID read as 74-3829516; printed 74-3829156. Provider 1 also misnamed "Bright Oak Preschool"	Extraction	Fidelity check: every value read must appear on its page	—	—

Also flagged (no change to tax, or decisions for the partner):

- A possible unfiled 2023 Form 3520 for Erik's ~150,000 Swedish inheritance. This was the prior preparer's open item. Advise the client in writing.

- The SEP excess grows to 4,326. It needs a corrective withdrawal by 10/15/2026, or a 6% excise on Form 5329.
- California AMT (Schedule P) on the 40,000 ISO spread is not computed by the engine, and most of the service period was in California. This is a blocking partner note.
- Two late 1099-NECs for contractors, and the FBAR confirmation. Both are outside the engagement.
- Electing §179 for the studio equipment saves 1,050 in California at no federal cost (engine what-if).
- An unreported Meridian wash sale (M12/M16) with zero net 2025 effect.
- The 2024 home office was overstated by 80 sq ft (400 of deduction). Immaterial; not amended.
- Three questionnaire answers are contradicted by documents (qa-14, qa-15, qa-31).

Independent recomputations that agree with the engine (16 of 16): W-2 sums, excess social security, the 8606 conversion, depreciation for both rentals, the condo gain, the S corp basis roll-forward using the 2023 K-1, SE tax, CTC phase-out, the SALT cap, Form 8938 values, Brightpath's 260 lots against the 1099-B summary, and every total against its inputs.

The timeline's 07/20/2026 "balance paid" event assumes a balance due. With these fixes the finalized return shows an overpayment, and the Filing page says so instead of computing interest on a balance that doesn't exist.

Technical decisions and tradeoffs

Decision	Why	Tradeoff
Packet kept immutable; all work in an event journal	Originals, audit trail, undo, versions, freeze and amend all fall out of one model; events never conflict, so several writers merge trivially	Every view replays the journal; fine at hundreds of events, a real system would snapshot
Permissions enforced at replay, not only in the UI	A forged or stale event can't change the return; refusals are visible evidence	Rules live in code (`ROLE_CAN`); a firm would want them as configurable policy

Decision	Why	Tradeoff
Sign-offs as content hashes per area	Reopening is automatic and explainable ("changed since signed")	Any change in the area reopens it, even a trivial one; a finer scope (per line) would be noisier
Checks as code over the packet	Each finding links to the exact evidence and proposes the adjustment	Written against this packet's document types; production would express them as declarative rules per form
Tax effect from per-dollar rates, exact what-ifs where available	Honest within the brief (the engine can't be rerun)	Linear: a short-term loss that offsets long-term gain is slightly misstated until the engine recomputes; downstream lines are flagged
Simulated documents behind a renderer interface	Highlights find each value's printed text; traces are stored as doc + page + value id	Replacing with page images only swaps `locate()` to use the position boxes
Preact + htm, no build framework, one static page + `packet.json`	Opens from a link anywhere; no server	No server-side enforcement; in production the journal append would be checked server-side with the same rules
Shared database when available, browser storage otherwise	Live multi-user inside claude.ai, still works offline	Presence is light and there is no locking; last write per event
Season clock filters facts and events by date	The whole season (late K-1, corrected 1099, reject, 2028 exam) can be replayed and demoed	Actions are stamped with the clock's date, which is right for a demo but not for production

How AI was used

I used Claude (Anthropic) as a pair engineer throughout. It worked inside a sandbox with the packet and the brief.

- **Analysis:** It read all 74 documents' printed content against the extracted values, the three engine versions, the agency records and the client answers. It also cross-checked them with short Python scripts. Each error in the table was confirmed against the printed page, not against the model's reading alone.
- **Build:** It wrote most of the code, and I directed and reviewed the design: the evidence graph, the journal, replay with permissions, and the checks. The checks were tested in Node against the packet, and the UI was tested in a headless browser for each role and each point in the season.
- **Writing:** It drafted this document and the video script.

[Edit this section to describe your own part accurately: what you decided, what you rewrote, and what you verified yourself.]

Video walkthrough script (about 9 minutes)

Record your screen and voice, for example with Loom. Before recording: open the app, go to Firm › "Reset workpaper journal", then "Load demo season activity". Set the clock to "Jul 9 · final review" and act as Priya.

Time	On screen	Say (in your own words)
0:00	Overview	"I didn't rebuild a spreadsheet. A workpaper exists to prove a number: where it came from, what was done to it, who checked it, what's open. So I built it as an evidence graph plus an append-only journal."
0:45	Overview: summary vs last year, refund tiles	"The engine's v3 says the client owes 4,220 federal. After the errors I found, it's a refund of about 4,680. California's refund goes from 8,256 to 11,627."

Time	On screen	Say (in your own words)
1:15	Findings, then open "RSU sales taxed with zero basis"	"These checks don't trust the engine. They read the printed documents, the IRS and FTB records, the client's answers and the prior year. This one is the biggest: noncovered RSU lots with zero basis."
2:00	Click an evidence chip (D11 p2)	"Every number traces to the page and the exact field. The documents are simulated from printed content, behind a renderer interface. Real PDFs would only change how a value is located on the page."
2:40	Return lines › Schedule C › SchC.1	"From the other direction: a line shows its rule, its inputs, its sources, its value in v1, v2 and v3, and the 2024 variance. The 1099-K was double-counted here, and that's also the item the IRS examines in 2028."
3:20	Schedules › Sales detail, tick "Exceptions only", select and mark agreed, click Undo in the toast	"Large detail: 295 lots tie to each broker's 1099-B, and the exceptions filter shrinks them to the ones that need eyes. Bulk actions are one batch, so one undo reverses them."
4:00	Apply F-THEO-SSN, then Adjustments	"Adjustments never overwrite. Each keeps the original, who, why, and the tax effect from the engine's per-dollar rates. Anything over 1,000 needs a partner."
4:40	Switch to Helen (partner), approve; try to approve her own; switch to Marcus and open D08	"Roles are enforced when the journal is replayed, not just by greying out buttons. Self-review is refused. Marcus has no clearance, so SSNs stay masked, and every unmasked view is logged."
5:30	Area page › sign-offs; Activity › refused events	"A sign-off is a hash of everything in its area. When v3 arrived in June, the March sign-offs reopened on their own."

Time	On screen	Say (in your own words)
6:10	Filing: blockers, Submit e-file refused	"Firm standards gate filing: open review notes, unapproved adjustments, stale approvals. The client's 8879 approval is fingerprinted too, so changing a number after approval reopens it."
6:50	Clock → Jul 13 · e-file rejected	"The season clock replays the packet's timeline. The e-file rejection is for Theo's SSN, which the workpaper had already flagged from the prior-year return."
7:20	Clock → Jan 2028 · Letter 566	"Freeze at filing, amend on top. The notice response is built from the frozen record, so it stands on its own years later."
7:50	Compare versions; Design notes	"Tradeoffs: tax effects are linear until the engine recomputes, so downstream lines are flagged rather than faked. The checks are code against this packet; in production they'd be declarative rules. The storage is a shared live database here, browser storage elsewhere."
8:40	Close	"What I'd do next: run the checks as rules per form type, enforce the journal server-side, and put real page images behind the same renderer."